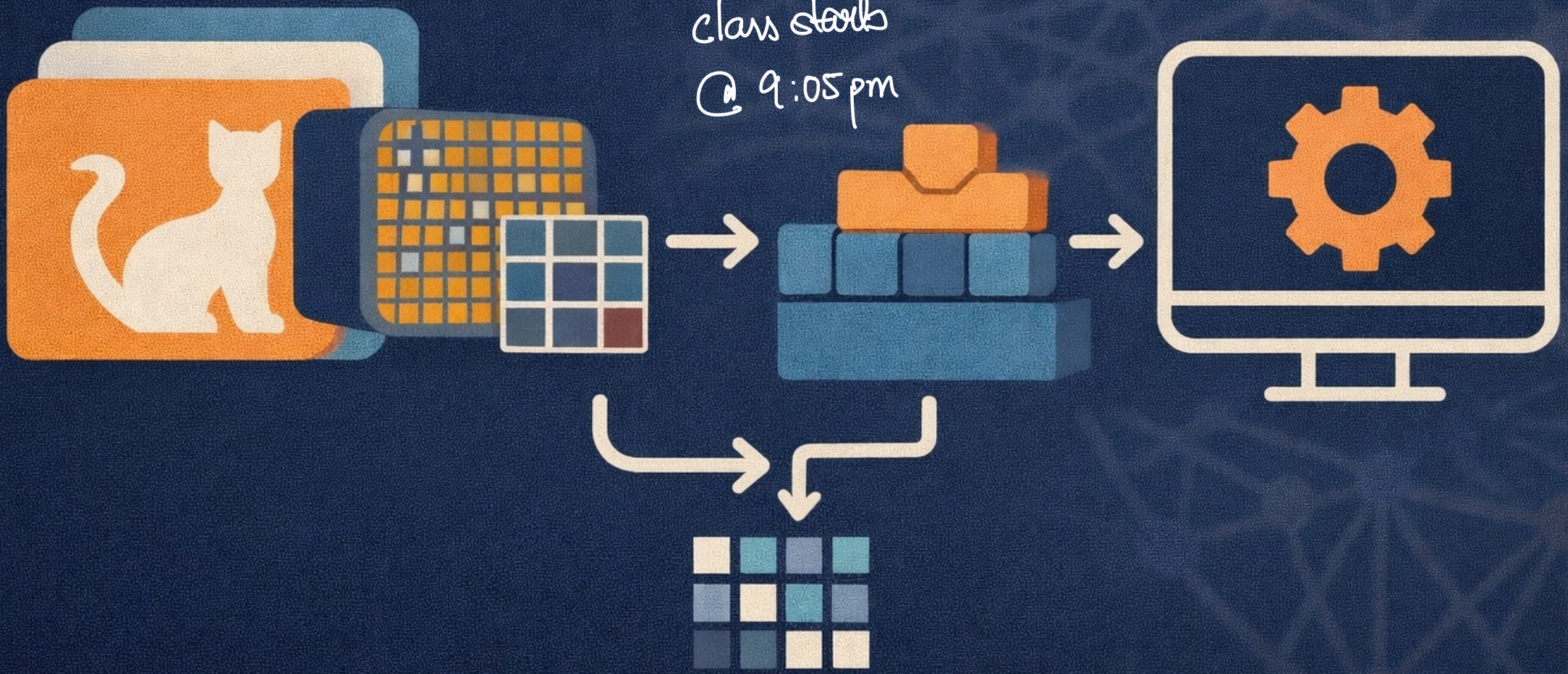
CONVOLUTIONAL NEURAL NETWORK



Lecture : Introduction to CV and CNN

↳ Convolutional Neural Network

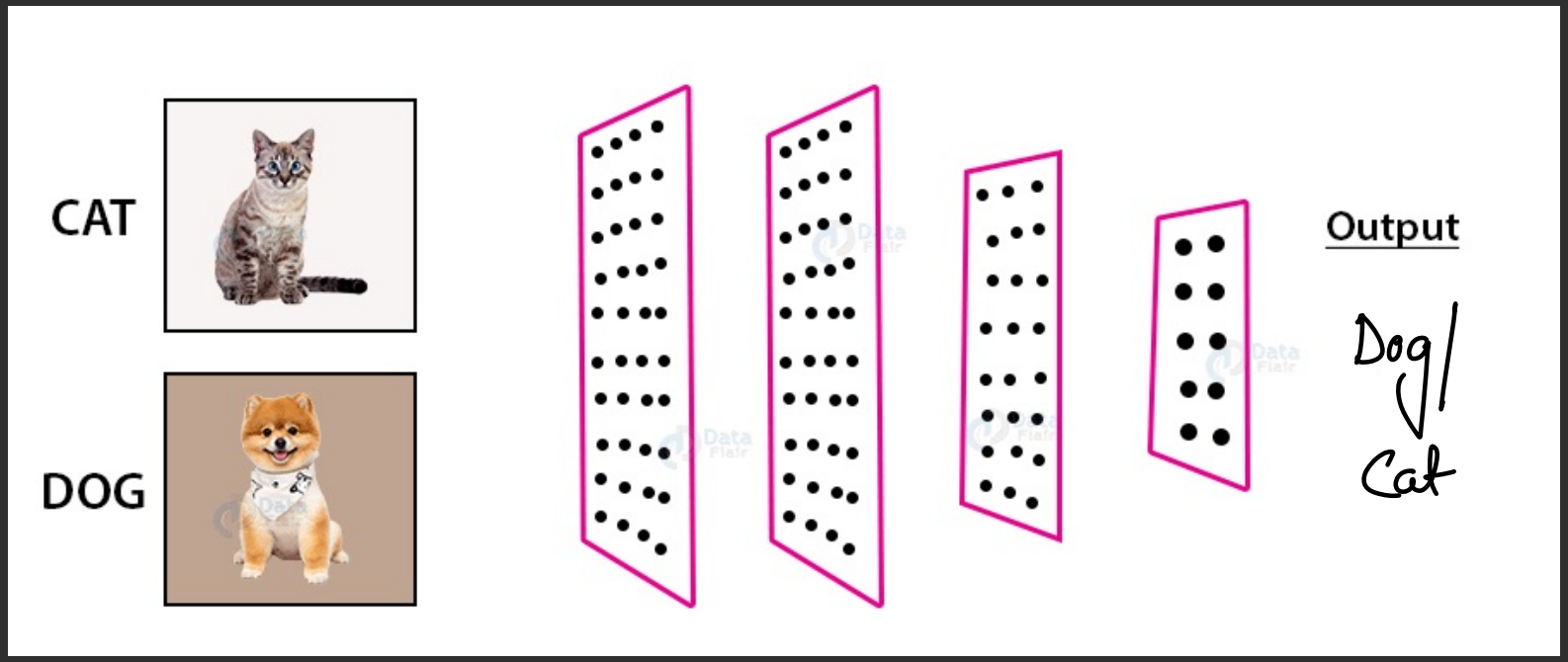
#Agenda

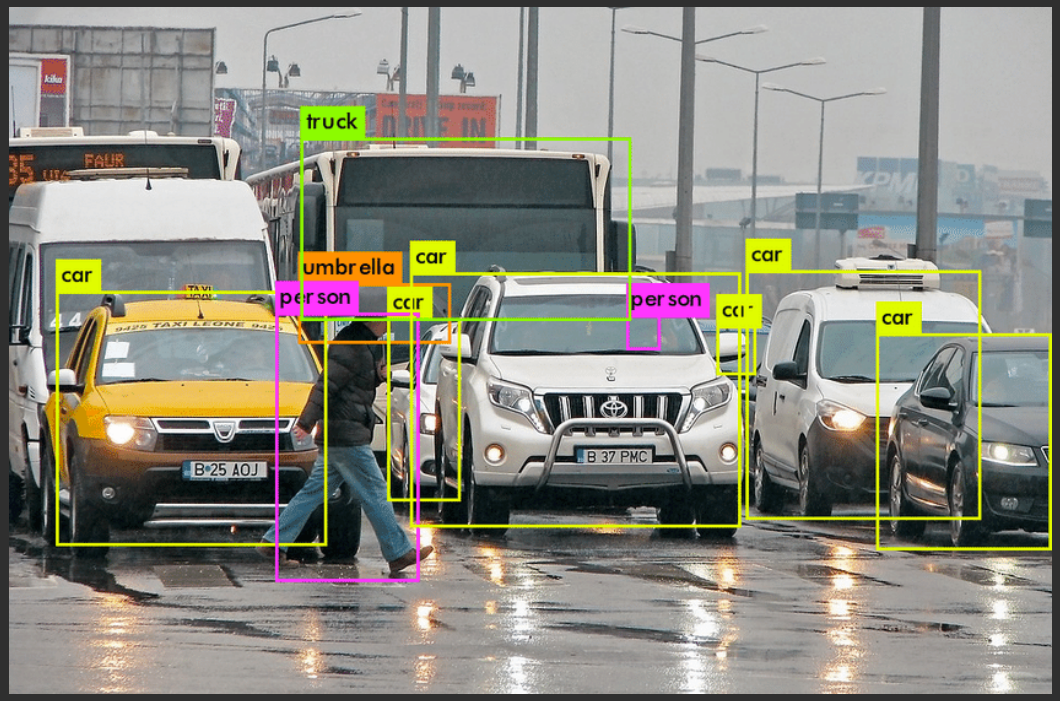
1) Computer Vision
2) Why CNN
3) Components of CNN

Computer Vision

A field of AI dealing with images

Image Classification





Object Detection



'50 40',
'बसरुरकर मार्किट
BASRURKAR MARKET',
'SPEED LIMIT'

Text Extraction from Image (OCR)

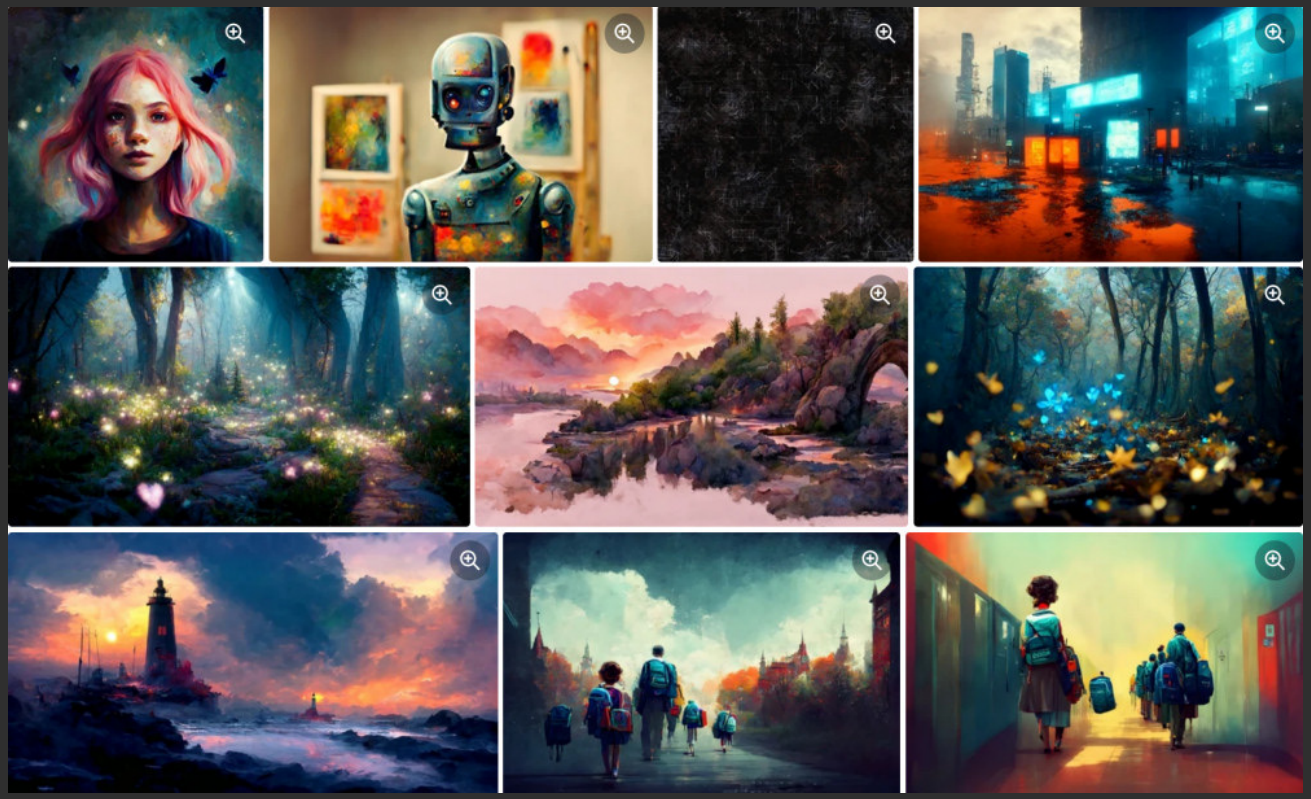
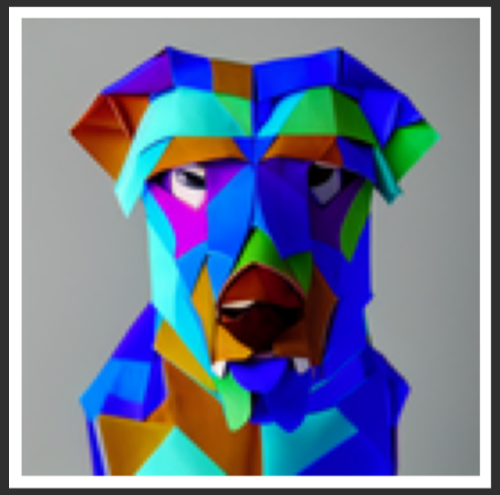


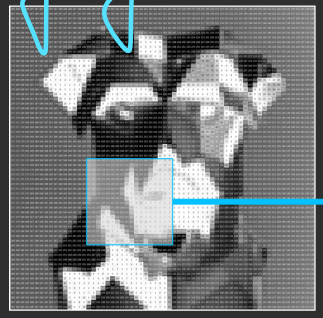
Image generation

What Computer Actually see?





grey scale

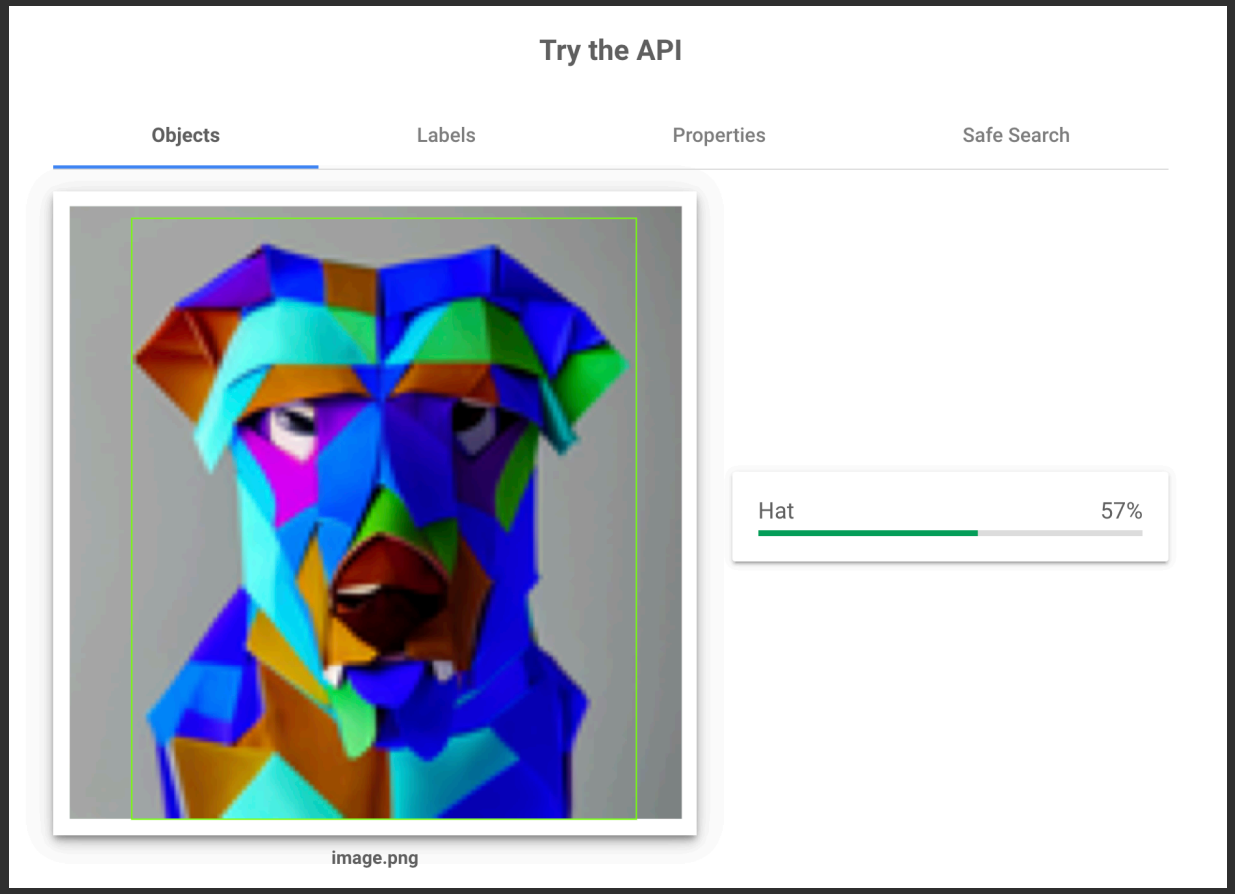


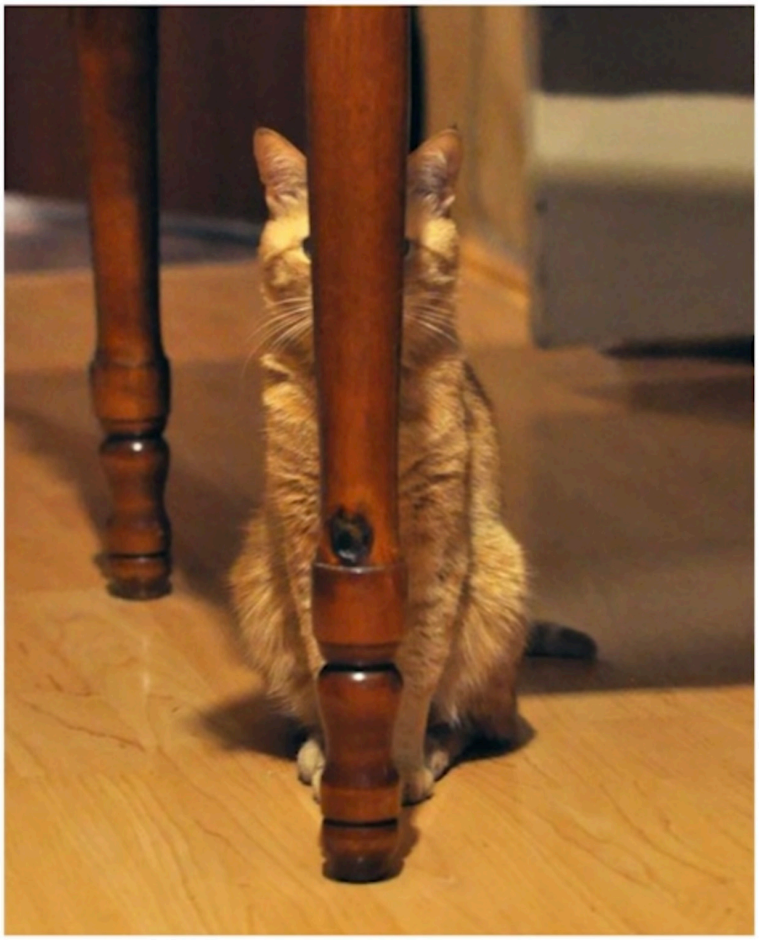
pixel

158	176	246	246	251	241	235	242	254	249	244	253	248	255	127	0
159	172	243	247	249	239	240	251	255	185	220	255	249	244	27	4
160	168	239	248	247	250	253	252	246	109	247	250	255	160	4	28
161	164	237	248	248	249	249	255	199	15	234	255	254	97	27	3
162	163	235	250	248	249	246	255	122	0	188	255	195	24	0	4
162	162	233	252	249	249	251	250	44	0	139	255	62	0	8	6
163	158	228	254	249	246	255	188	0	0	93	185	0	0	0	0
161	165	236	252	249	246	255	190	0	0	38	68	13	50	78	87
160	224	253	247	249	248	249	251	58	0	12	25	55	86	100	67
207	255	251	249	255	247	247	255	189	0	8	32	0	0	0	0
255	251	255	145	144	255	244	248	253	58	0	7	12	12	9	5
255	248	251	46	0	192	255	241	255	112	0	3	1	3	3	3
248	255	205	3	0	22	229	250	255	167	0	8	1	4	3	2
243	255	154	0	12	0	66	251	253	209	5	12	10	5	5	3
245	255	182	16	0	7	0	116	255	232	30	0	3	5	1	5
250	252	227	155	25	7	2	0	169	255	57	8	34	4	1	4

28 x 28 = 784 pixels

#Why CV is hard?





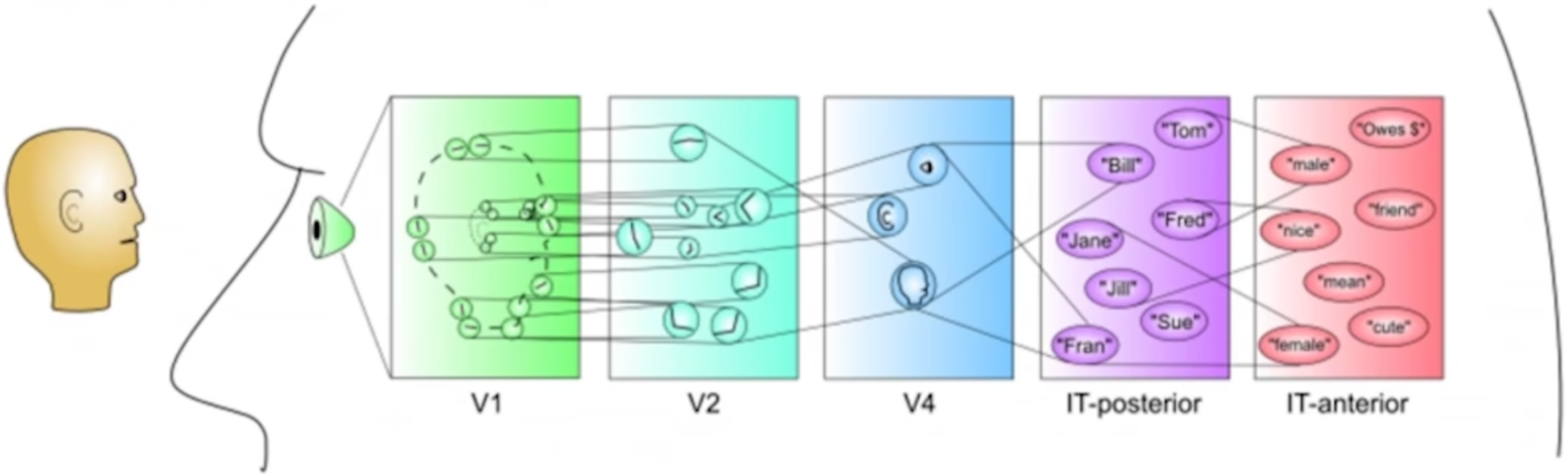


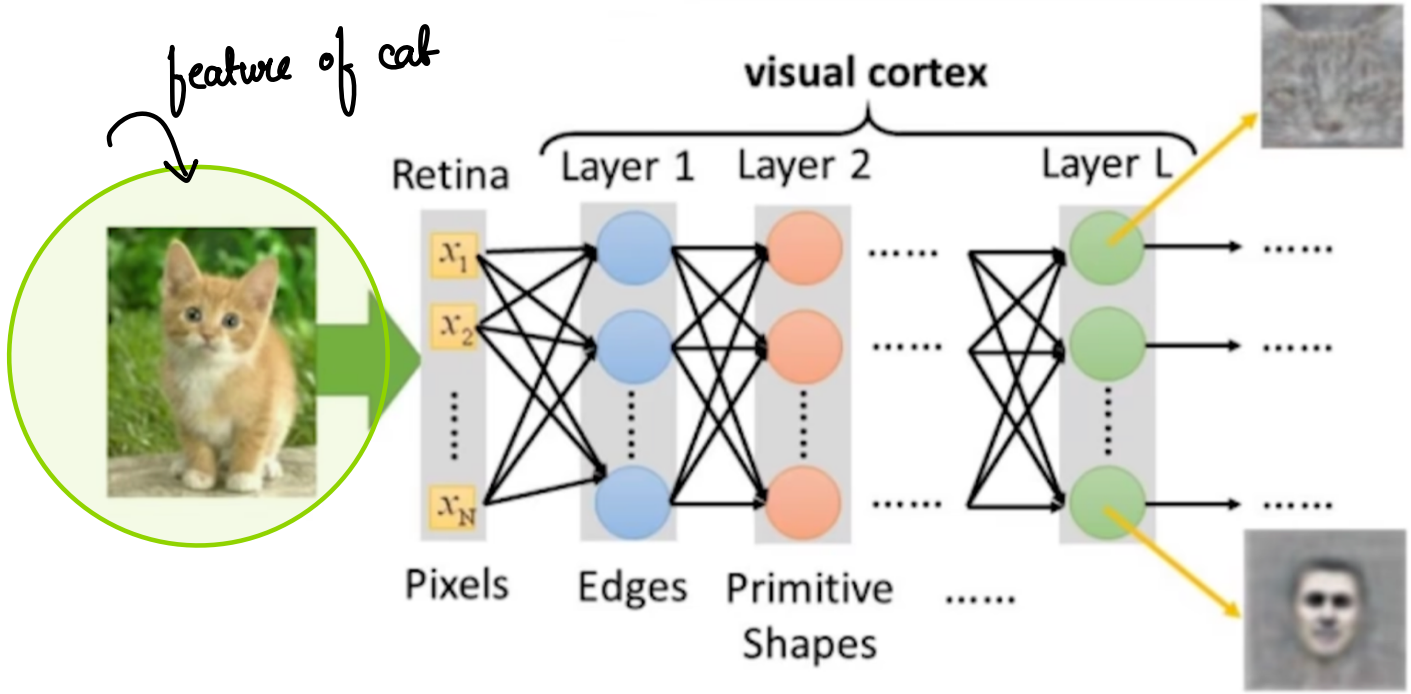
✓ ANN → Artifical Neural N/w

✓ CNN → Convonutional Neural N/w

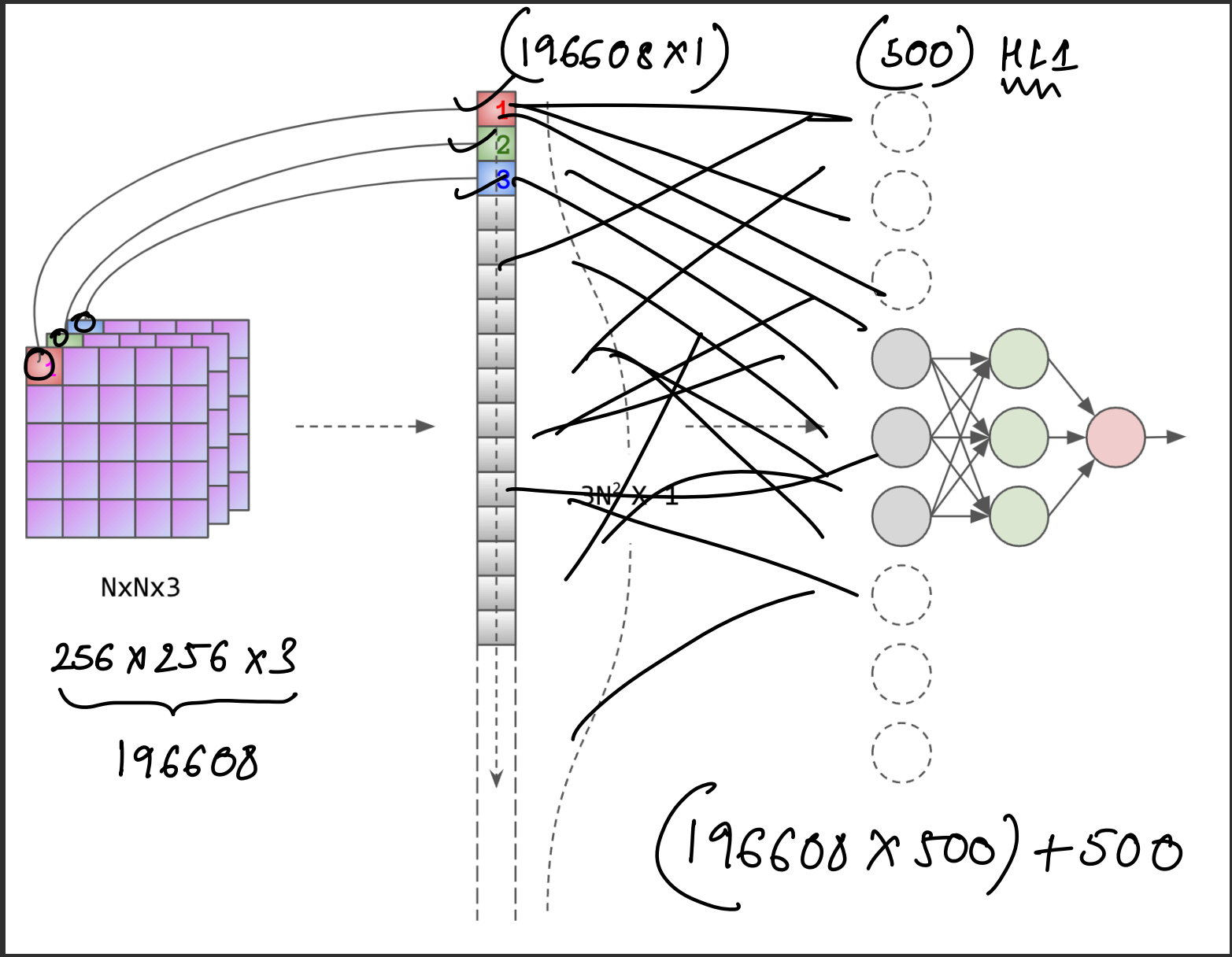
Visual Cortex

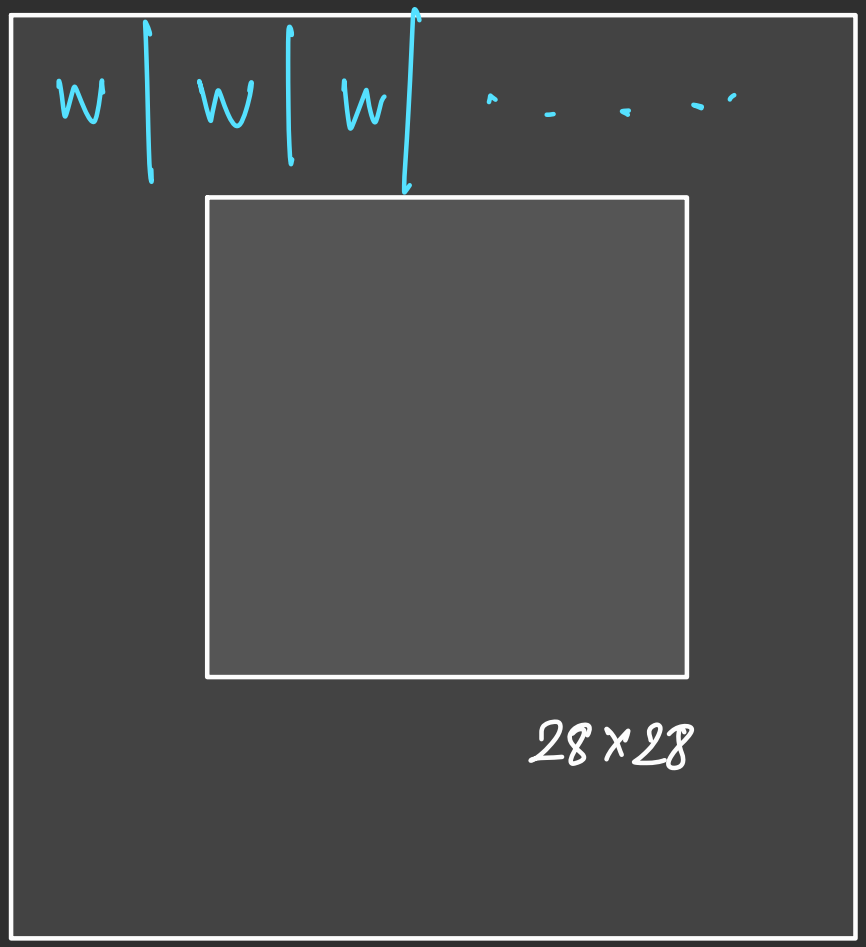
(Its Structure is Instructive and Inspiring)

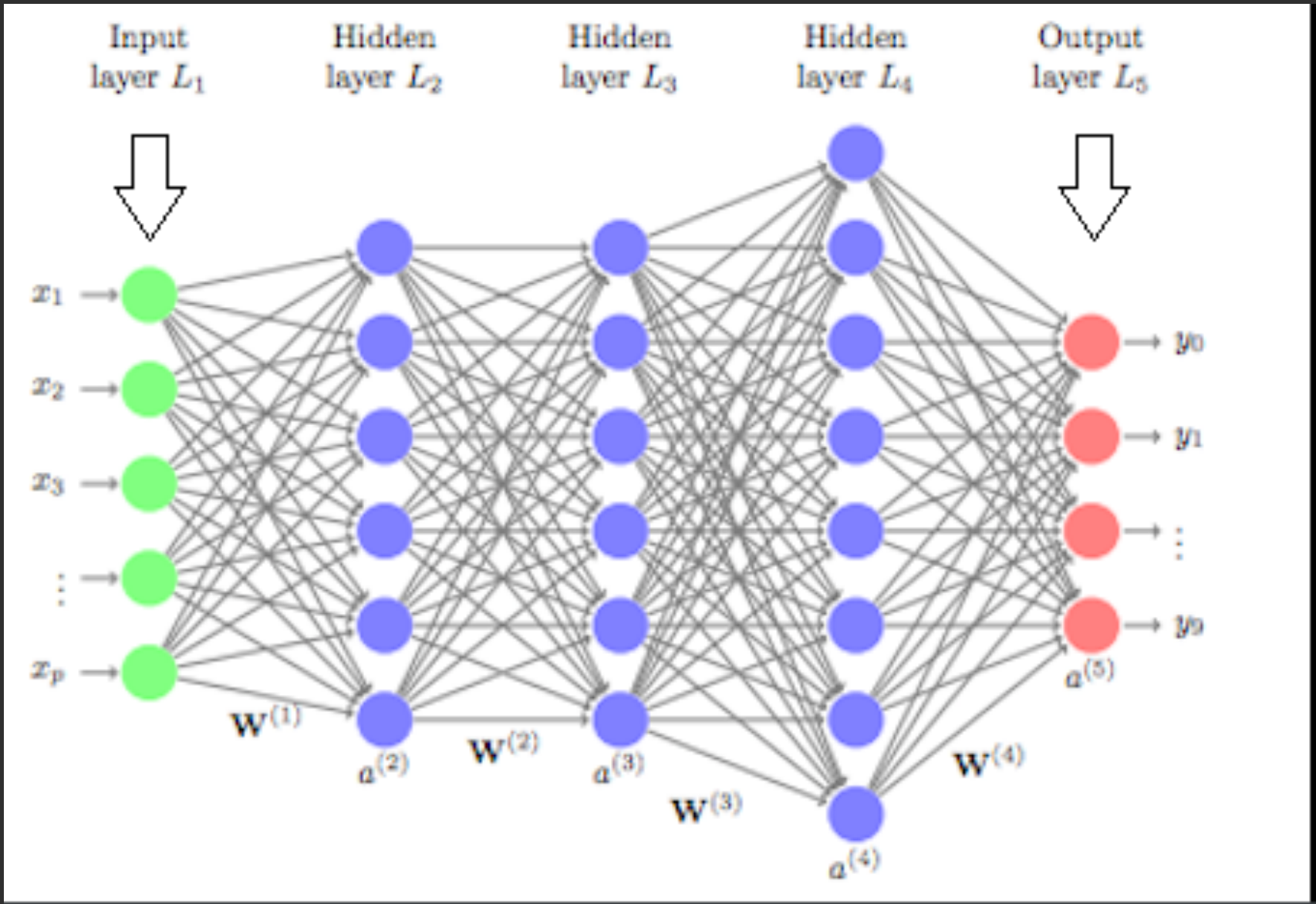




Using ANN


(196608 x 1)
(500) HL1
1
2
3
$3N^2$ X 1
NxNx3
256 x 256 x 3
196608
(196608 x 500) + 500


w
w
w
28×28


Input layer L_1
Hidden layer L_2
Hidden layer L_3
Hidden layer L_4
Output layer L_5
x_1
x_2
x_3
x_p
y_0
y_1
y_9
$\mathbf{W}^{(1)}$
$\mathbf{W}^{(2)}$
$\mathbf{W}^{(3)}$
$\mathbf{W}^{(4)}$
$a^{(2)}$
$a^{(3)}$
$a^{(4)}$
$a^{(5)}$



128 x 128





256 x 256

Target	T_1	T_2	T_3	T_4
3	0	0	1	0
2	0	1	0	0
4	0	0	0	1
1	1	0	0	0

4 class



#Why Use CNN?

1) High Computational Cost

2) Overfitting

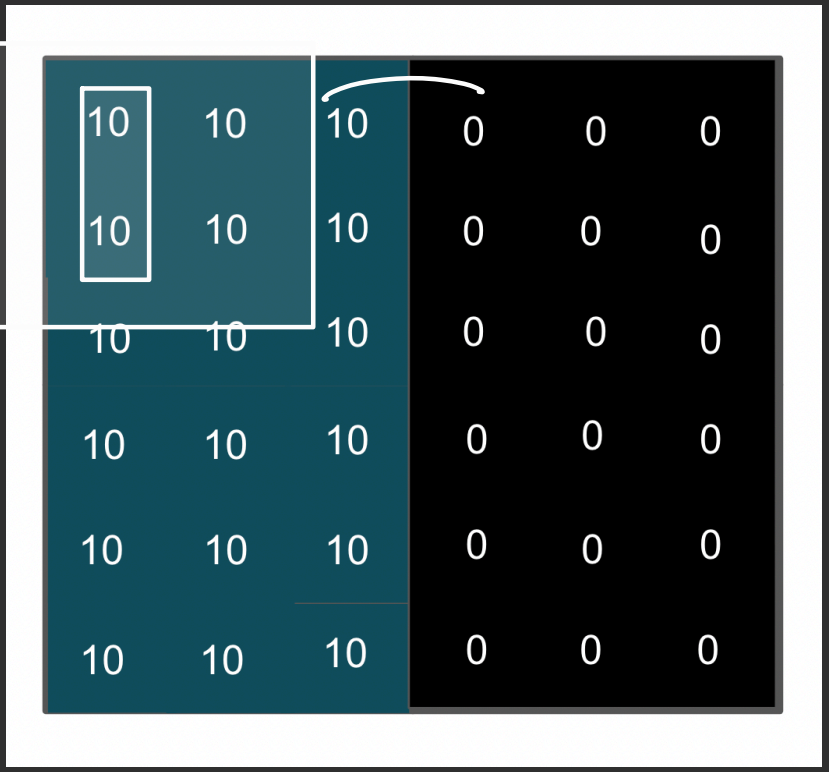
3) Loss of important info like spatial arrangements of pixels.

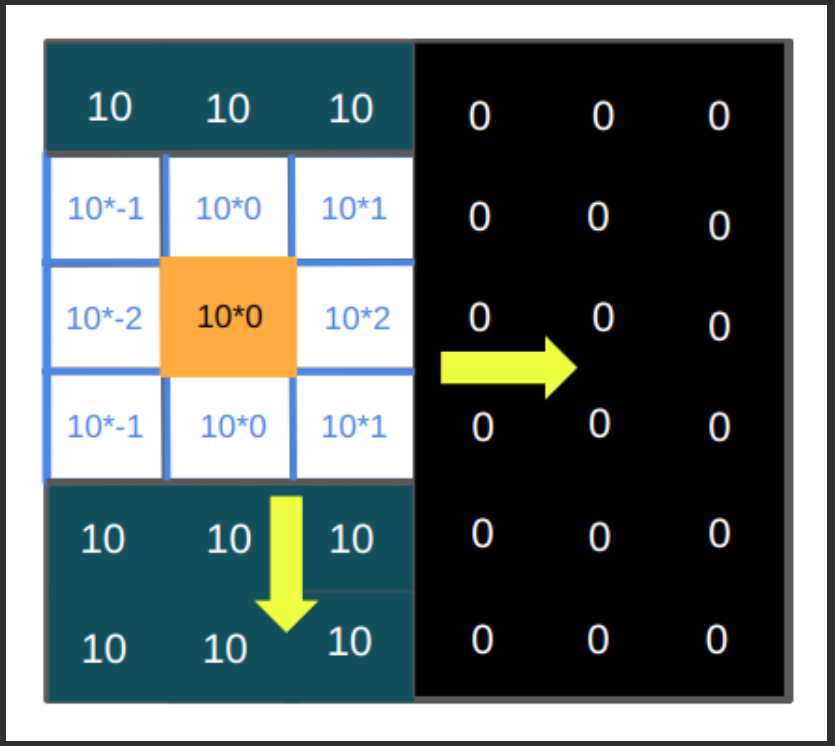
1024x1024x3

"Huge features"



What is Convolution?





> Kernal / Filter



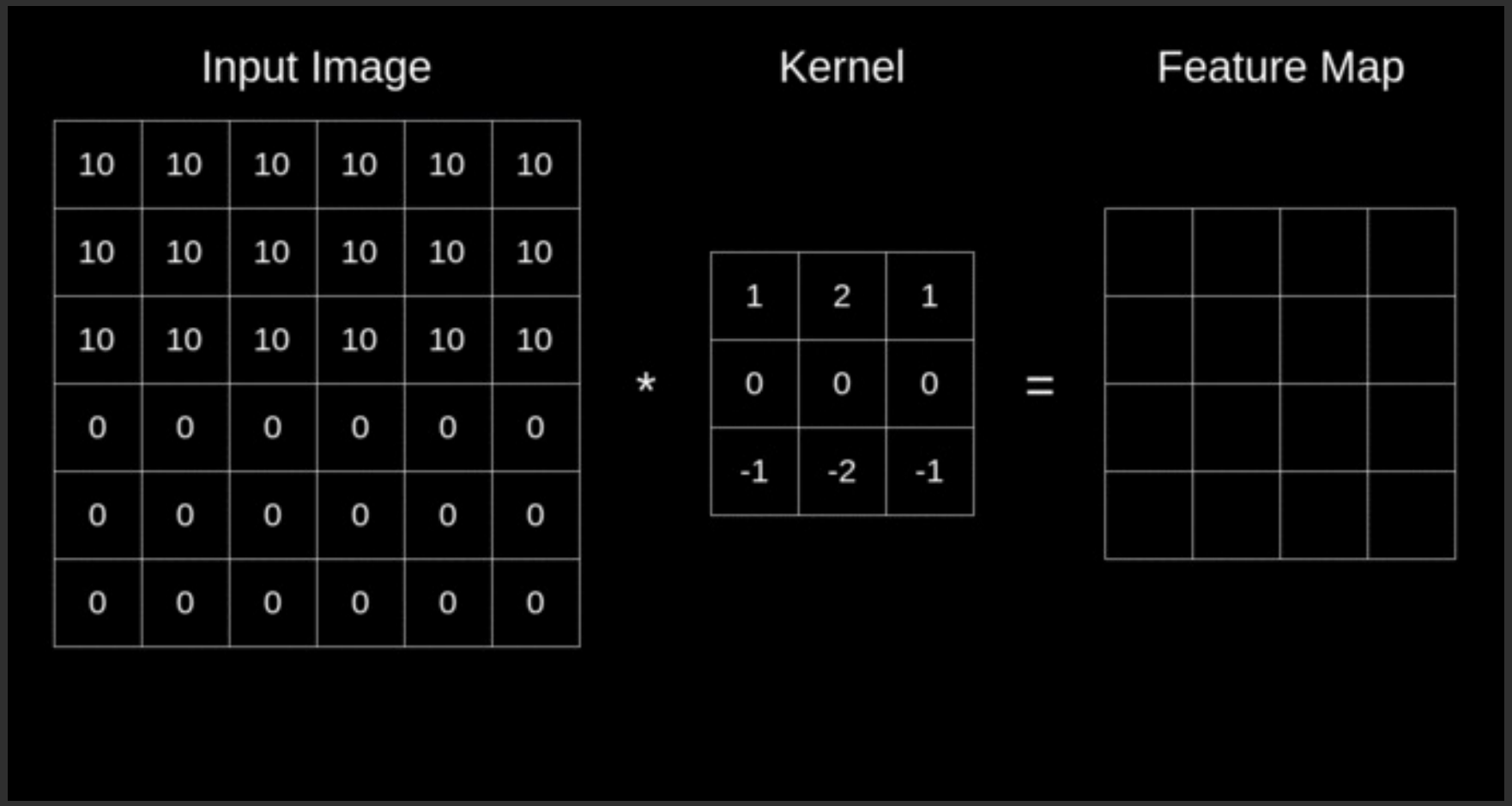








vertical edges


Input Image
Kernel
Feature Map
10 10 10 10 10 10
10 10 10 10 10 10
10 10 10 10 10 10
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
*
1 2 1
0 0 0
-1 -2 -1
=

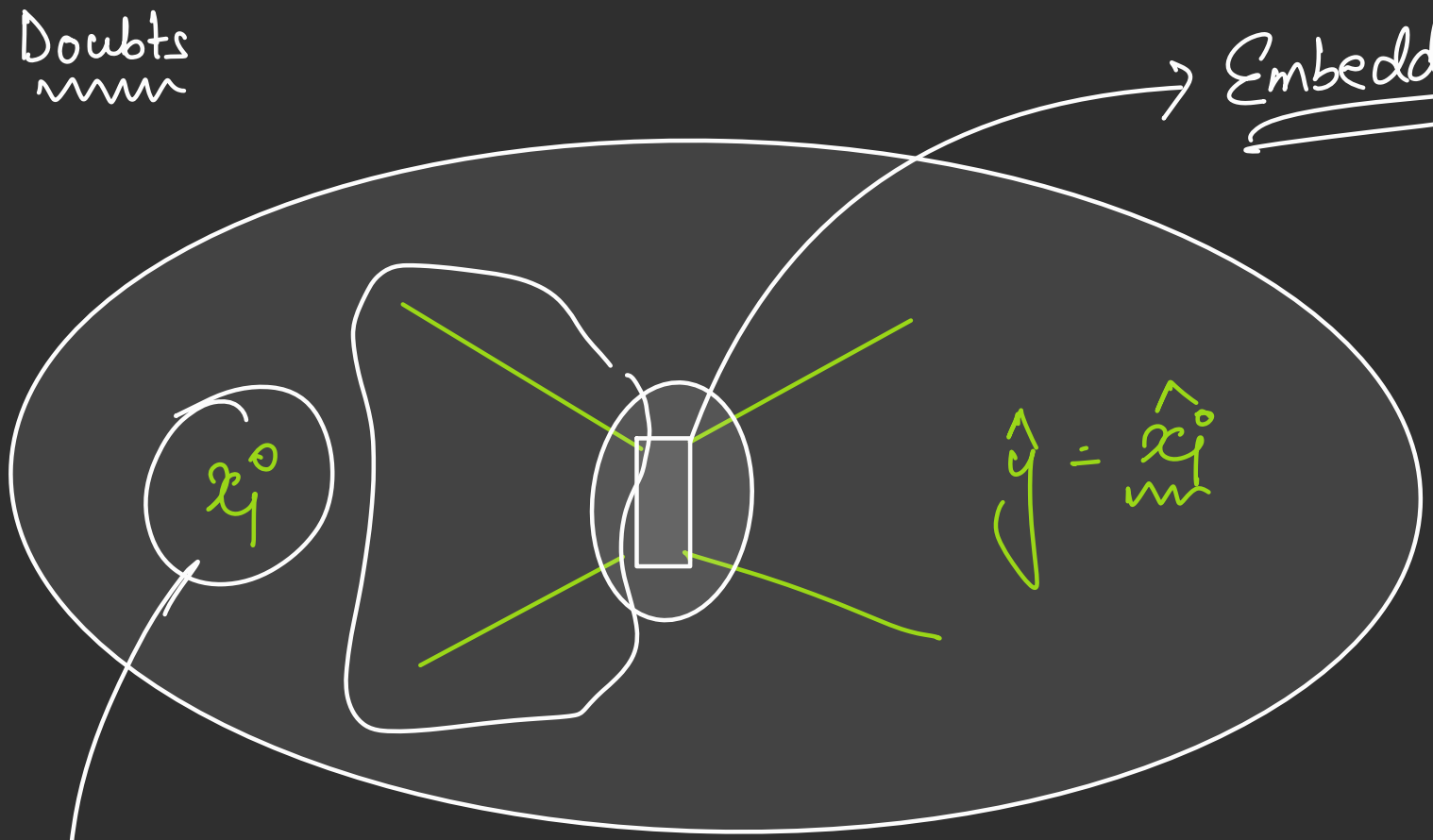
H Kernal

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

V Kernal

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

Doubts



Auto Encode

HD

Data

$x_i^0 \longrightarrow$ Encoder $\longrightarrow$ Embedde